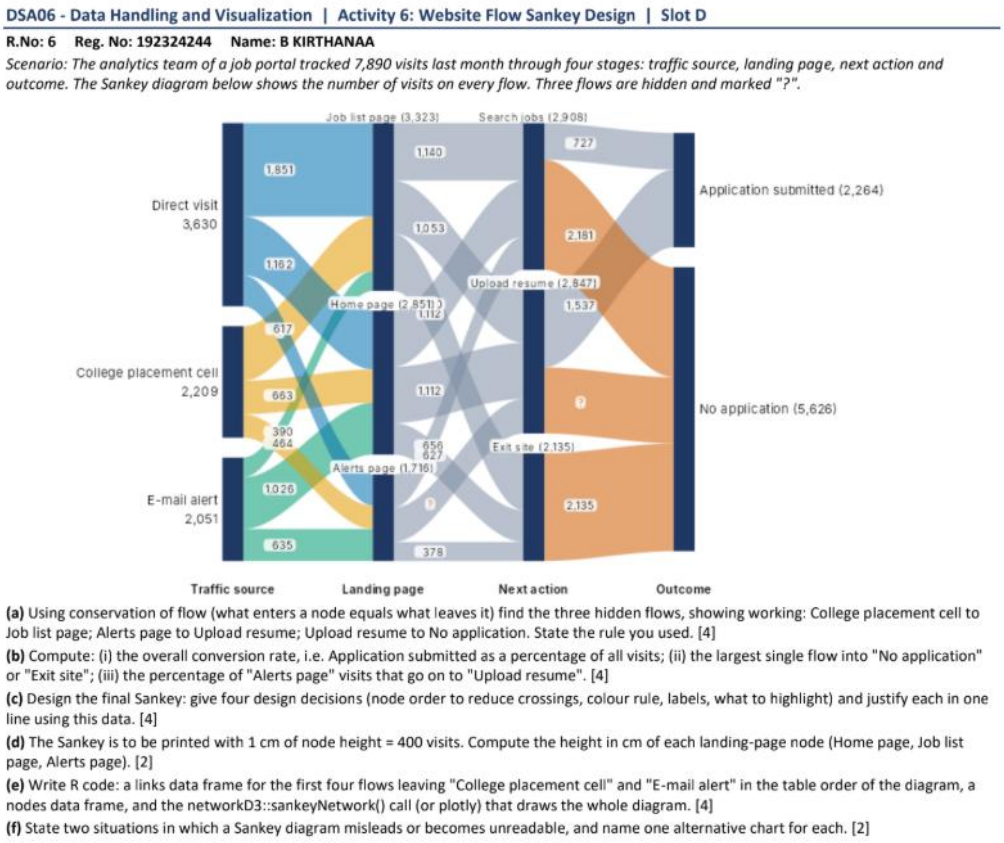
DSA06 - Data Handling and Visualization

Activity 6: Website Flow Sankey Design

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Scenario: The analytics team of a job portal tracked 7,890 visits last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of visits on every flow. Three flows are hidden and marked "?".



Given Sankey diagram (three flows hidden and marked "?")

Rule used (conservation of flow): what enters a node equals what leaves it. For every landing-page and next-action node, total inflow = node total = total outflow. The traffic-source nodes only send out visits, the outcome nodes only receive them, and every column adds up to 7,890 visits.

PART (a): Recovering the three hidden flows

Using conservation of flow, find the three hidden flows, showing working: College placement cell to Job list page; Alerts page to Upload resume; Upload resume to No application. State the rule you used.

Hidden flow	Working (outflow of the source node)	Cross-check (inflow of the target node)	Answer (visits)
College placement cell → Job list page	2,209 − 663 (Home) − 464 (Alerts) = 1,082	3,323 − 1,851 (Direct) − 390 (E-mail) = 1,082	**1,082**
Alerts page → Upload resume	1,716 − 656 (Search jobs) − 378 (Exit site) = 682	2,847 − 1,053 (Job list) − 1,112 (Home) = 682	**682**
Upload resume → No application	2,847 − 1,537 (Application submitted) = 1,310	5,626 − 2,181 (Search jobs) − 2,135 (Exit site) = 1,310	**1,310**

Ans:

The hidden flows are 1,082 (College placement cell → Job list page), 682 (Alerts page → Upload resume) and 1,310 (Upload resume → No application). The rule used is conservation of flow: inflow = node total = outflow. Both methods give the same number for each flow, which confirms the answers.

PART (b): Conversion measures

Compute: (i) the overall conversion rate, i.e. Application submitted as a percentage of all visits; (ii) the largest single flow into "No application" or "Exit site"; (iii) the percentage of "Alerts page" visits that go on to "Upload resume".

Measure	Working	Answer
(i) Overall conversion rate	Application submitted ÷ all visits × 100 = 2,264 ÷ 7,890 × 100	**28.7%**
(ii) Largest single flow into "No application" or "Exit site"	Into No application: Search jobs 2,181; Exit site 2,135; Upload resume 1,310 (recovered). Into Exit site: Job list page 1,130; Home page 627; Alerts page 378. The largest of all is Search jobs → No application, which is 2,181 ÷ 5,626 = 38.8% of all non-applying visits.	**2,181 visits** **(Search jobs → No application)**
(iii) Alerts page visits going to Upload resume	Alerts → Upload resume ÷ Alerts page visits × 100 = 682 ÷ 1,716 × 100	**39.7%**

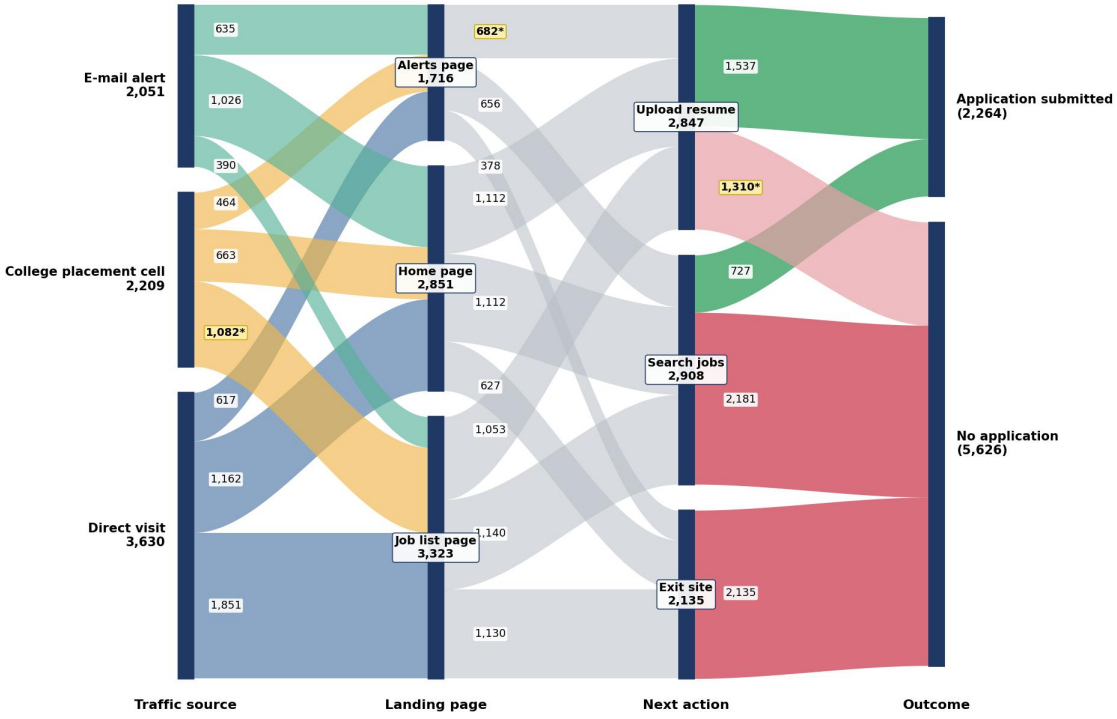
Ans:

(i) 28.7% of all visits end in an application. (ii) The largest flow into No application is Search jobs → No application (2,181 visits), narrowly ahead of Exit site (2,135). The largest flow into Exit site is Job list page → Exit site (1,130), so 2,181 is the largest of the two. (iii) About 39.7% of Alerts page visits go on to Upload resume.

PART (c): Design of the final Sankey

Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data.

Decision	Choice	Justification (one line)
1. Node order	Traffic source: E-mail alert, College placement cell, Direct visit. Landing page: Alerts, Home, Job list. Next action: Upload resume, Search jobs, Exit site. Outcome: Application submitted, No application.	I checked all 216 possible orders with Application submitted on top. None gets below 19 crossing pairs, but this order has the least crossing between large flows (about 27% less than the given order). It also puts successful paths at the top and drop-offs at the bottom.
2. Colour rule	First-stage flows take their traffic-source colour (Direct blue, College amber, E-mail green). Middle flows are light grey. Flows into Application submitted are green, flows into No application are red.	Colour shows origin first and result last, so good and bad outcomes can be told apart without a legend, and the grey keeps the busy middle quiet.
3. Labels	Node name with total visits on every node, the visit count on every flow (thousands separators), and stage headings. No "?" is left; recovered flows are marked *.	The reader can check the conservation of flow and does not have to guess values from flow width.
4. Highlight	The two largest drop-offs, Search jobs → No application (2,181) and Exit site → No application (2,135), in strong red; the third (1,310) is faded.	Together they are 4,316 visits (54.7% of all visits), the biggest chance to improve conversion, so the main message is visible at first glance.



Output 1: Final Sankey design (hidden flows recovered, * = recovered flow)

PART (d): Node heights for printing

The Sankey is to be printed with 1 cm of node height = 400 visits. Compute the height in cm of each landing-page node (Home page, Job list page, Alerts page).

Landing-page node	Visits	Height = visits ÷ 400	Height
Home page	2,851	2,851 ÷ 400 = 7.1275	**7.13 cm**
Job list page	3,323	3,323 ÷ 400 = 8.3075	**8.31 cm**
Alerts page	1,716	1,716 ÷ 400 = 4.29	**4.29 cm**

Ans:

Home page = 7.13 cm, Job list page = 8.31 cm, Alerts page = 4.29 cm. Check: together they are 19.73 cm, which equals 7,890 ÷ 400 (without the gaps between nodes).

PART (e): R code with network D3

Write R code: a links data frame for the first four flows leaving "College placement cell" and "E-mail alert" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call that draws the whole diagram.

```
library(networkD3)

# Nodes data frame: one row per node, in the final design order (top to bottom)
nodes <- data.frame(
  name = c("E-mail alert", "College placement cell", "Direct visit",
           "Alerts page", "Home page", "Job list page",
           "Upload resume", "Search jobs", "Exit site",
           "Application submitted", "No application"),
  group = "node"
)

# Links data frame: first four flows leaving College placement cell and E-mail alert
links_first4 <- data.frame(
  source = c("College placement cell", "College placement cell",
             "College placement cell", "E-mail alert"),
  target = c("Job list page", "Home page", "Alerts page", "Job list page"),
```

```
  value  = c(1082, 663, 464, 390),
  group  = c("college", "college", "college", "email")
)

# Remaining 19 flows, so that the whole diagram is drawn
links_rest <- data.frame(
  source = c(rep("E-mail alert", 2), rep("Direct visit", 3),
             rep("Job list page", 3), rep("Home page", 3), rep("Alerts page", 3),
             rep("Search jobs", 2), rep("Upload resume", 2), "Exit site"),
  target = c("Home page", "Alerts page",
             "Job list page", "Home page", "Alerts page",
             "Search jobs", "Upload resume", "Exit site",
             "Search jobs", "Upload resume", "Exit site",
             "Search jobs", "Upload resume", "Exit site",
             "Application submitted", "No application",
             "Application submitted", "No application",
             "No application"),
  value  = c(1026, 635,
             1851, 1162, 617,
             1140, 1053, 1130,
             1112, 1112, 627,
             656, 682, 378,
             727, 2181, 1537, 1310, 2135),
  group  = c("email", "email", rep("direct", 3), rep("route", 9),
             "apply", "leak", "apply", "drop", "leak")
)

links <- rbind(links_first4, links_rest)

# sankeyNetwork() needs zero-based row numbers of the nodes
links$source <- match(links$source, nodes$name) - 1
links$target <- match(links$target, nodes$name) - 1

# Check: visits reaching the two outcomes must equal all visits
sum(links$value[links$target %in% c(9, 10)])   # 7890

# Colour rule: source colour, grey routing, green = applied, red = dropped
palette <- 'd3.scaleOrdinal()
  .domain(["node", "direct", "college", "email", "route", "apply", "drop", "leak"])
  .range(["#1F3864", "#4E79A7", "#F2B84B", "#59B39A", "#B8BEC8",
          "#2E9E5B", "#E9A0A9", "#D1495B"])'

sankeyNetwork(Links = links, Nodes = nodes,
              Source = "source", Target = "target", Value = "value",
              NodeID = "name", NodeGroup = "group", LinkGroup = "group",
              colourScale = palette, units = "visits",
              fontSize = 13, nodeWidth = 20, nodePadding = 20,
              iterations = 0, sinksRight = FALSE)
```

Explanation of the R code

- **nodes:** one row for each of the 11 boxes. The row order is the top-to-bottom order of the final design.
- **links_first4:** the first four flows leaving College placement cell (Job list 1,082, Home 663, Alerts 464) and E-mail alert (Job list 390), in the table order of the diagram.
- **links_rest:** the other 19 flows. Together with links_first4 this gives all 23 flows, so the whole diagram is drawn.
- **match(...) - 1:** sankeyNetwork() expects the source and target as zero-based row numbers of nodes, so the names are converted to numbers.
- **palette / group:** applies the colour rule from part (c) through the group column.
- **iterations = 0, sinksRight = FALSE:** keeps the node order given in the nodes table instead of letting the layout re-order it.

Output: the call draws the diagram shown in Output 1, with all four stages and the 23 flows.

PART (f): When a Sankey misleads

State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each.

Situation	Why it misleads or becomes unreadable	Alternative chart
1. Too many nodes and many small flows (for example every page of the portal)	Flows cross each other like spaghetti, the thin flows cannot be read or compared by width, and labels overlap.	Heatmap of a source × destination table (or a bar chart of the top flows)
2. Movement that is not one-way (users go back to an earlier page, or loop between pages)	A Sankey only shows forward movement from stage to stage; going back has to be drawn as duplicate nodes, which hides the repeat visits and gives a wrong picture of the journey.	Network graph or chord diagram

Summary

All 23 flows after recovering the hidden values (shaded = hidden flow recovered in part (a)):

From	To	Visits	From	To	Visits
Direct visit	Job list page	1,851	Home page	Search jobs	1,112
Direct visit	Home page	1,162	Home page	Upload resume	1,112
Direct visit	Alerts page	617	Home page	Exit site	627
College placement cell	Job list page	**1,082***	Alerts page	Search jobs	656
College placement cell	Home page	663	Alerts page	Upload resume	**682***
College placement cell	Alerts page	464	Alerts page	Exit site	378
E-mail alert	Job list page	390	Search jobs	Application submitted	727
E-mail alert	Home page	1,026	Search jobs	No application	2,181
E-mail alert	Alerts page	635	Upload resume	Application submitted	1,537
Job list page	Search jobs	1,140	Upload resume	No application	**1,310***
Job list page	Upload resume	1,053	Exit site	No application	2,135
Job list page	Exit site	1,130†			

** recovered in part (a). † not clearly labelled in the diagram; 3,323 − 1,140 − 1,053 = 1,130.*

Measure	Value
Total visits (every column adds up to this)	7,890
Overall conversion rate (Application submitted)	2,264 ÷ 7,890 = 28.7%
Best converting next action	Upload resume: 1,537 ÷ 2,847 = 54.0% apply
Alerts page visits going to Upload resume	682 ÷ 1,716 = 39.7%

Conclusion

Conservation of flow rebuilds every hidden value in a Sankey diagram, and each result can be checked from both the outflow and the inflow side. Only 28.7% of visits end in an application, and most of the loss comes from Search jobs and Exit site. Visitors who reach Upload resume convert best (54.0%), so the portal should push more visits to that page.